EX. NO: 01

DATE: 22/07/2026

8 QUEENS PROBLEM

AIM:

To place 8 queens on an 8x8 chessboard so that no two queens each other, i.e., no two queens share the same row, column, or diagonal.

PYTHON CODE:

```
def is_safe(board, row, col):
    for i in range(col):
        if board[row][i] == 1:
            return False

    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    i, j = row, col
    while i < 8 and j >= 0:
        if board[i][j] == 1:
            return False
        i += 1
        j -= 1

    return True


def solve_queens(board, col):
    if col >= 8:
        return True

    for row in range(8):
        if is_safe(board, row, col):
            board[row][col] = 1
```

```
            if solve_queens(board, col + 1):
                return True

            board[row][col] = 0

    return False


def print_board(board):
    print("8-Queens Solution:\n")
    for row in board:
        for cell in row:
            if cell == 1:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()


board = [[0 for _ in range(8)] for _ in range(8)]

if solve_queens(board, 0):
    print_board(board)
else:
    print("No solution exists.")
```

OUTPUT:

```
8-Queens Solution:

Q . . . . . . .
. . . . . . Q .
. . . . Q . . .
. . . . . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . Q . . . . .
```

RESULT:

Thus the program of placing 8 queens on an 8x8 chessboard was executed successfully.